MADHUMITHA S

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate (B.E., CGPA 7.80) with hands-on skills in embedded systems, IoT, and microcontroller programming in Embedded C. Experienced in interfacing sensors, actuators, displays, and Wi-Fi modules on Arduino and ESP32 through an Embedded IoT internship and hardware projects, with a complementary background in UI/UX design using Figma.

TECHNICAL SKILLS

Languages: Embedded C, C, Arduino C/C++
Microcontrollers & Boards: Arduino Uno / Nano, ESP32, ESP8266 (NodeMCU)
Embedded Concepts: GPIO, ADC, PWM, Timers, Interrupts, UART, I2C, SPI, Debouncing, Sensor Calibration
IoT & Connectivity: Wi-Fi, Bluetooth, MQTT, HTTP, ThingSpeak, Blynk IoT, Cloud Data Logging
Sensors & Modules: DHT11, HC-SR04 Ultrasonic, PIR, MQ-2 / MQ-135 Gas, Soil Moisture, Relay, L298N Motor Driver, 16x2 LCD (I2C), 4x4 Keypad
Tools: Arduino IDE, Tinkercad Circuits, Proteus, Serial Monitor / Plotter
Design: UI/UX Design, Figma, Canva, Adobe Express
Soft Skills: Communication, Teamwork, Leadership, Adaptability

EXPERIENCE

Embedded Internet of Things (EIoT) Internship
Embedded IoT Intern

- Worked with microcontrollers, sensors, and embedded systems, gaining practical exposure to IoT concepts from sensing to cloud connectivity.
- Interfaced digital and analog sensors with microcontroller boards and programmed them in Embedded C, reading data over GPIO and ADC.

PROJECTS

Password-Based Motor Control System *Arduino, Embedded C, Keypad, LCD, Relay*

- Developed a secure motor control system in which only users entering the correct password on a 4x4 keypad can switch the motor ON/OFF through a relay driver.
- Added a 16x2 LCD for user prompts and status, with lockout after three wrong attempts to block unauthorised access.

IoT Smart Home Automation *ESP32, Wi-Fi, Blynk IoT, Relay Module*

- Built an ESP32-based system to control four home appliances from a smartphone over Wi-Fi using a 4-channel relay module and the Blynk IoT app.
- Kept manual switch control as a fallback, so appliances remain usable when the internet connection drops.

IoT Environment Monitoring Station *ESP32, DHT11, MQ-135, ThingSpeak*

- Designed a station that measures temperature, humidity, and air quality and uploads readings to the ThingSpeak cloud every 15 seconds for live graphs.
- Implemented threshold-based alerts with an on-board buzzer and LED when gas levels or temperature exceed safe limits.

Smart Irrigation System *Arduino Uno, Soil Moisture Sensor, Relay, Pump*

- Automated plant watering by switching a water pump ON only when soil moisture falls below a calibrated threshold, and OFF once the soil is adequately wet.
- Removed the need for manual watering and avoided over-watering by irrigating based on actual soil condition rather than a fixed schedule.

EDUCATION

Vivekanandha College of Technology for Women — B.E., Electronics & Communication Engineering 2023 – 2027
CGPA: 7.80 / 10.0 | Tiruchengode, Namakkal, Tamil Nadu

Vignesh International School (CBSE) — Higher Secondary (HSC) 2023
Percentage: 62%

The Path Global Public School (CBSE) — Secondary School (SSLC) 2021
Percentage: 68%

ACHIEVEMENTS & CERTIFICATIONS

- NPTEL — Introduction to Internet of Things (IoT)
- Embedded IoT Internship Certification
- UI/UX Design Certification
- Typewriting (English Junior) — First Class